

Original article

Phase II study of docetaxel and cisplatin in patients with recurrent or disseminated squamous-cell carcinoma of the head and neck

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Summary

Background: Results with docetaxel as single drug in squamous-cell head and neck cancer have been encouraging. The purpose of the present phase II study is to evaluate the antitumour efficacy and toxicity of the combination of docetaxel and cisplatin in patients with recurrent or disseminated squamous-cell carcinoma of the head and neck (SCCHN) for whom no curative therapy is available.

Patients and methods: Eligibility criteria included: written informed consent; WHO performance status ≤ 2 ; age 18–70 years; adequate bone marrow, liver, and renal function; measurable or evaluable disease; no previous systemic chemotherapy (prior radiotherapy and/or surgery were allowed); no other previous or concurrent malignancy; no peripheral neuropathy. Treatment consisted of docetaxel 75 mg/m² in a one-hour infusion after pre-treatment with prednisolone, followed by cisplatin 75 mg/m² in a half-hour infusion preceded and followed by hydration. Treatment was repeated every three weeks for a maximum of eight cycles.

Results: Twenty-five patients (median age 52 years, range 33–66) entered the trial, all were evaluable for survival, twenty-four for response and toxicity. Twenty-four patients had undergone prior radiotherapy and seventeen had also had surgery. Nineteen had local-regional recurrence only, three had local-

regional disease and distant metastases, and three had distant metastases only. Patients received a median of 5 treatment cycles (range 2–8). Overall response rate was 33% (8 of 24) of patients; complete response rate was 8% (2 of 24) of patients, lasting 2.2 and 17.1 months, respectively; partial response rate was 25% (6 of 24) of patients, lasting for a median of 4.9 months (range 1.7–11.6 months). Median survival was 11 months. Toxicity was relatively well tolerated. However, one patient died of probable toxicity (neutropenia and infection) and three patients discontinued treatment because of toxicity (massive oedema, myocardial infarction, persistent thrombocytopenia). The most frequent moderate-to-severe toxicity (75% of patients) was grade 3–4 neutropenia, transient in all but one patient. Grade 3 neuropathy occurred in one patient, none had grade 4. Grade 3 oral mucositis occurred in three patients, none had grade 4. Grade 2–3 hypomagnesaemia occurred in 10 patients requiring magnesium infusion.

Conclusions: Docetaxel and cisplatin is an active combination in patients with recurrent or disseminated SCCHN. Remissions are however fairly short. Toxicity is significant, but generally manageable.

Key words: cisplatin, disseminated disease, docetaxel, head and neck cancer, recurrent disease, squamous-cell carcinoma

Introduction

Squamous-cell carcinoma of the head and neck (SCCHN) remains local-regional for a long time and distant metastasis is usually a late manifestation. SCCHN is therefore curable in a significant proportion of patients by local-regional therapy, i.e., radiotherapy and/or surgery. However, nearly 50% of primary cases recur after aggressive standard therapy, and prognosis for recurrent or metastatic disease is poor. Cisplatin and 5-fluorouracil (5-FU) is the most commonly used combination chemotherapy regimen for recurrent or metastatic SCCHN, but overall response rates have only been around 30% with few complete responses and a median survival of only four to six months [1–4]. No chemotherapy regimen has yet shown any curative potential in this group of patients. However, because of the devastating consequences of the relentless local-regional progression of

incurable SCCHN, the achievement of tumour regression and remission lasting for some time is important for the patient's well-being and quality of life in the remaining weeks and months of his/her life [5].

Results of phase II studies of docetaxel as single drug in SCCHN seemed encouraging, with response rates of 21%–42% [6–8]. Its principal toxicity was transient myelosuppression [6–8]. Combinations of docetaxel and cisplatin had only been tried in phase I studies, but results in SCCHN seemed promising.

Based on these findings the present phase II study was initiated with the purpose of evaluating the efficacy and toxicity of the combination of docetaxel and cisplatin in patients with recurrent or disseminated SCCHN for whom no curative therapy was possible. Based on toxicity data from phase I studies of the combination of docetaxel and cisplatin in SCCHN and other solid tumours [9] the schedule for the present study was fixed

at 75 mg/m² for both drugs given at three-week intervals.

Patients and methods

Patient eligibility

Patient eligibility criteria included histologically confirmed SCCHN that recurred after previous radiotherapy and/or surgery with no possibility of further curative treatment, or which presented with distant metastases at the time of initial diagnosis; measurable (i.e., bidimensionally measurable) or evaluable (i.e., unidimensionally measurable) disease; performance status 2 or less according to WHO criteria; age 18–70 years; adequate bone marrow, liver, and renal function; no symptoms of peripheral neuropathy; no previous systemic chemotherapy; no other previous or concurrent malignancy (except adequately treated carcinoma *in situ* of the cervix uteri or basal cell carcinoma of the skin); no severe intercurrent illness; and signed informed consent. The protocol for this study was approved by the Ethics Committees of Copenhagen, Denmark.

Treatment plan

Within two weeks of study entry, all patients had a complete history and physical examination, complete blood count, serum chemistries (liver and renal function tests and electrolytes), ⁵¹Cr-EDTA clearance, ECG, chest radiograph, and if indicated computed tomographic (CT) scans of the head and neck or other relevant areas for later response evaluation. Patients were monitored during treatment with a weekly complete blood count.

Treatment consisted of docetaxel 75 mg/m² in a one-hour infusion followed by cisplatin 75 mg/m² in a half-hour infusion. The shorter infusion time for cisplatin compared with the phase I study [9] was chosen to conform with the usual way of administering cisplatin in our department. Treatment was repeated every three weeks for a maximum of eight cycles.

All patients received oral prednisolone 50 mg 12 hours and 1 hour before docetaxel and b.i.d. for 4 days following docetaxel infusion. Cisplatin infusion was preceded by i.v. hydration with 1000 ml saline and 500 ml mannitol 20% and was followed by 2000 ml saline i.v.

Successive cycles of chemotherapy were given every three weeks for a maximum of eight cycles if the patient's blood count had returned to normal and nonhaematologic toxic effects > grade 2 had resolved. The dose of docetaxel was reduced by 25% if the granulocyte nadir in the preceding cycle was < 0.5 × 10⁹/l for > 7 days and/or associated with fever (≥ 38.5 °C), if the thrombocyte nadir in the preceding cycle was < 25 × 10⁹/l, or if non-haematologic side effects ≥ grade 2 had occurred in the preceding cycle. The dose of cisplatin was reduced by 25% if the ⁵¹Cr-EDTA clearance was reduced to 50%–75% of the normal value, and cisplatin would have been discontinued if even greater reductions in ⁵¹Cr-EDTA clearance had occurred. Once the dose of one of the drugs had been decreased due to toxicity no attempt at re-escalation was attempted.

Follow-up history and physical examination, tumour measurement and side-effect assessment were performed before each cycle. Pertinent radiographs and CT scans were repeated after every two cycles to determine response.

Response criteria

Criteria of complete response, partial response, no change, and progressive disease were based on the standard definitions of the WHO [10]. Duration of response was calculated from the time of first documentation of objective response (complete or partial) to first documentation of disease progression. This fairly conservative way of calculating duration of response (as opposed to calculating from start

of chemotherapy) was chosen because it represents the most relevant measure of the time the patient experiences relief of the symptoms caused by the tumour. Responses were assessed after every two cycles of treatment.

Survival evaluation

Survival was calculated from start of chemotherapy until death, surviving patients were censored at the date of last follow-up. Survival curves were constructed by the method of Kaplan and Meier [11].

Toxicity evaluation

Toxicity grading was based on the National Cancer Institute (NCI) Common Toxicity Criteria [12]. Patients were evaluated weekly for haematological toxicity and every three weeks (i.e., before each new treatment cycle) for nonhaematologic toxicity.

Results

Patient characteristics

Patients were entered into this study from November 1995 to June 1998. Twenty-five patients were assessable for survival, twenty-four for response and toxicity. Patient characteristics are listed in Table 1. At entry into the present study all but one of the patients had been previously treated with radiotherapy alone or radiotherapy and surgery. Radiotherapy had in all cases been given with curative intent to large fields to a total dose of 60–70 Gy in 2 Gy fractions. The single previously untreated patient presented with distant metastases to bone and bone marrow in addition to his primary nasopharyngeal carcinoma (poorly differentiated SCC) with neck node metastases. Nineteen patients had local-regional recurrence only, three had both local-regional disease and distant metastases, and three had distant metastases only. All patients who failed local-regionally did so within or partly within previous radiation fields. Patients received a median of 5 cycles of treatment (range 2–8).

Response and survival results

Complete remission was achieved in two patients (8%), lasting 2.2 and 17.1 months, respectively. Partial remission was achieved in six patients (25%) with a median duration of 4.9 months (range 1.7–11.6 months). The overall response rate was thus 33%, and the median duration of these responses was 4.9 months (95% confidence interval (95% CI) 2.2–7.7 months). Figure 1 shows remission duration from time of remission for both complete and partial responders.

Table 2 shows details regarding response in relation to disease localisation at study entry. Four of eighteen evaluable patients with local-regional disease only had objective responses (2 complete, 2 partial); two of three patients with both local-regional disease and distant metastases had objective responses (both partial); and

Table 1. Patient characteristics.

Characteristics	Number of patients (%)
Total	25
Age (in years)	
Median	52
Range	33–66
Sex	
Male	21 (84)
Female	4 (16)
Performance status	
0	12 (48)
1	10 (40)
2	3 (12)
Primary disease site	
Oral cavity	5 (20)
Nose	1 (4)
Nasopharynx	3 (12)
Oropharynx	7 (28)
Hypopharynx	5 (20)
Larynx	4 (16)
Prior therapy	
None	1 (4)
Radiotherapy alone	7 (28)
Radiotherapy + surgery	17 (68)
Disease status at study entry	
Local-regional	19 (76)
Local-regional + distant metastasis	3 (12)
Distant metastasis only	3 (12)
Distant metastasis, total	6
Lung	4
Bone and bone marrow	1
Subcutaneous	1
Number of cycles of treatment	
2	10
3	1
4	1
5	2
6	4
7	2
8	5

Table 2. Response according to sites of recurrence.

Disease site at entry	Number of patients	Number of CR	Number of PR	Total number of responses
Local-regional only	18	2	2	4
Local-regional + distant metastasis	3	0	2	2
Lung	2	0	1	
Bone and bone marrow	1	0	1	
Distant metastasis only	3	0	2	2
Lung	2	0	2	
Subcutaneous	1	0	0	
Total	24	2	6	8

Abbreviations: CR – complete response; PR – partial response.

two of three patients with distant metastases only had objective responses (both partial). Responses were seen even in heavily irradiated sites (2 complete, 3 partial). However, patients with distant metastases with or without local-regional disease seemed to respond more

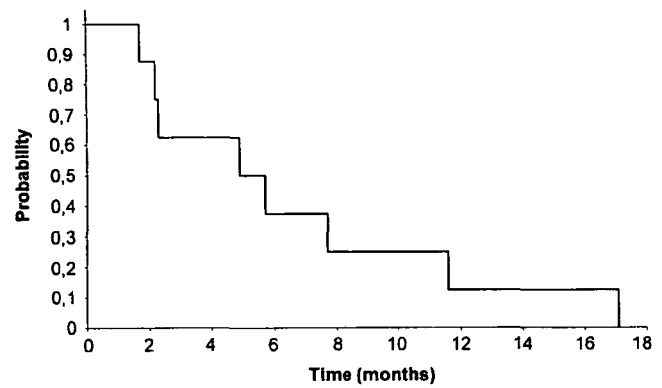


Figure 1. Remission duration from time of remission for complete and partial responders.

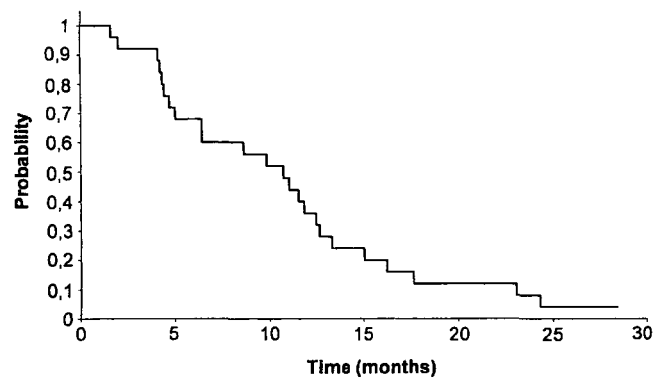


Figure 2. Overall survival from start of chemotherapy.

frequently than patients with local-regional recurrence only, although the numbers are too small for any meaningful statistical analysis.

Overall survival for all 25 patients is shown in Figure 2. Median survival was 11 months (95% CI: 6.4–12.4 months). One- and two-year survival rates were estimated to be 36% and 4%, respectively. Twenty-four deaths have occurred to date among the twenty-five patients, one patient is alive with progressive disease at 28.4 months.

Toxic effects

Toxic effects in each patient are listed in Table 3, and toxic effects in each treatment cycle are listed in Table 4.

After a total of 115 courses administered to the 25 patients toxicity was significant with 1 patient dying of probable toxicity (neutropenia and infection) after 3 courses. The most frequent moderate-to-severe toxicity was grade 3–4 neutropenia, occurring in 18 of 24 evaluable patients and in 35 of 115 treatment cycles. It was, however, transient except in the patient mentioned above who died of infection. In only three instances did the neutropenia last for more than seven days. One patient developed persistent low platelet count around $50 \times 10^9/l$ with no clinical bleeding, and he had to discontinue treatment. The median haematologic nadir counts in the 115 treatment cycles were: Haemoglobin 6.6 mmol/l (range 5.4–8.4), granulocytes $1.5 \times 10^9/l$

Table 3. Common toxic effects in 24 patients (NCI–Common Toxicity Criteria, highest grade for each patient).

	Grade 1	Grade 2	Grade 3	Grade 4
Neutropenia	2	1	8	10
Thrombocytopenia	6	1	0	1
Neutropenic fever/infection	1	5	1	1
Stomatitis/mucositis	9	4	3	0
Diarrhoea	10	5	1	0
Hypomagnesaemia	2	5	5	0
Hypersensitivity	2	0	0	0
Skin/nail changes	5	1	0	0
Alopecia	9	13	–	–
Neuropathy	10	1	1	0

(range 0.0–11.0), and platelets $205 \times 10^9/l$ (range 20–591).

Most patients experienced some nausea, vomiting and hair loss. Two patients experienced mild hypersensitivity reactions, which subsided after treatment with corticosteroids and antihistamines. A low magnesium occurred frequently with 10 patients developing grade 2–3 hypomagnesaemia requiring intravenous magnesium infusion. We have now started adding magnesium to the intravenous hydration fluids given before and after cisplatin, which may reduce this problem in the future. Mild neuropathy was frequent, but no patient developed moderate-to-severe neuropathy except for one patient who developed grade 3 hearing loss. Mucositis was not a great problem, despite the fact that most of the patients had previously received high-dose radiotherapy to significant parts of their oral and pharyngeal mucosa.

Treatment was delayed in 7 of 90 cycles, in 5 instances this was due to the patient's personal decision, in 1 instance due to alcohol intoxication, and in 1 instance due to poor general condition (this patient was the one who died of probable toxicity). Doses of docetaxel and cisplatin were reduced according to the criteria outlined above. The dose of docetaxel was reduced in 6 of 25 patients (24 of 115 cycles, in 21 cycles to 75%, in 3 cycles to 60%), the dose reduction was due to stomatitis in 1 patient (7 cycles), to neutropenic fever in 4 patients (15 cycles), to low granulocyte nadir without fever in 1 patient (1 cycle), and to oedema in 1 patient (1 cycle). The dose of cisplatin was reduced in 7 of 25 patients (23 of 115 cycles, in 15 cycles to 75%, in 8 cycles to 50%), the dose reduction was due to a decrease in ^{51}Cr -EDTA clearance (to values between 50% and 75% of normal) in 5 patients (18 cycles), to nausea in 2 patients (4 cycles), and finally the dose was erroneously reduced in 1 patient (1 cycle).

During the treatment period performance status increased in 5 patients, was unchanged in 15 patients, whereas it decreased in 5 patients (in 3 related to disease progression, in 1 related to toxicity which probably caused the patient's death, whereas 1 patient chose to stop treatment after 7 cycles and remained in complete remission for 1.3 years after the last cycle).

Three patients had to discontinue treatment because

Table 4. Total number of cycles with toxicity among 115 cycles (NCI–Common Toxicity Criteria)

	Grade 1	Grade 2	Grade 3	Grade 4
Neutropenia	11	18	19	16
Thrombocytopenia	20	1	0	1
Neutropenic fever/infection	1	12	2	1
Nausea/vomiting	15	22	4	0
Stomatitis/mucositis	22	7	4	0
Diarrhea	22	10	1	0

of toxicity. One patient with persistent low platelet count is mentioned above. One patient had massive oedema which did not respond to diuretics. This patient was, however, the only patient with significant oedema problems. He had received a total dose of 640 mg (337 mg/m^2) of docetaxel. Finally, one patient discontinued treatment because of a myocardial infarction. This patient was not seen at our institutions after the incident and was thus not evaluable for response or further toxicity; however, the patient is known to have survived almost two years after the incident.

Discussion

The combination of docetaxel and cisplatin showed marked activity in this phase II trial in recurrent or disseminated SCCN with a 33% objective response rate, 8% complete response rate, and a median response duration of 4.9 months. The response rate is similar to the results of the EORTC study of docetaxel 100 mg/m^2 and cisplatin 75 mg/m^2 , in which the response rate for pre-treated patients was also 33% [13]. Somewhat similar results have been reported by Forastiere et al. [14], whereas response rates have been higher in patients who were not previously treated [13, 15–17]. Response duration in pre-treated patients has not been clearly stated, but judging from the median survival [13] the response duration in the present study seems fairly long, especially as it was calculated in a very conservative way.

The toxicity of docetaxel and cisplatin was considerable. Most patients experienced some nausea, vomiting, and hair loss. The most frequent moderate-to-severe toxicity (in 75% of patients) was neutropenia, which was generally short-lived. However, one patient died of infection during a neutropenic episode. Sensory neuropathy and mucositis were not prominent. This toxicity profile is in accordance with other studies [13–16]. One frequent problem in our study was significant hypomagnesaemia requiring intravenous magnesium infusion in 42% of patients. This problem has not been reported by others in head and neck cancer patients, and it may possibly be forestalled by adding magnesium to the intravenous hydration fluids. One patient in our study had to discontinue treatment because of massive intractable oedema. This fluid retention syndrome induced by docetaxel is well known and appears to be caused by

capillary protein leakage [18]. In earlier studies the fluid retention syndrome was much more common [7]. However, the prophylactic administration of corticosteroids, as given in the present study, has markedly reduced the risk of this complication [8, 13, 19]. The prophylactic corticosteroids probably also explain the low frequency and low severity of hypersensitivity reactions [19].

The aim of treating patients with incurable SCCHN with chemotherapy is to achieve palliation. In this respect one might consider in future phase II studies to include quality of life questionnaires. However, it is notoriously difficult to achieve a meaningful statistical analysis of quality of life questionnaires in very small patient materials.

The two-drug regimen of docetaxel and cisplatin tested in the present study showed marked activity in this previously treated population, with response rates of the same order of magnitude as in the rather few other similar studies [13, 14]. The duration of response in the present study compares favourably with results of combination regimens not containing taxoids [1–4], although it must be borne in mind that comparisons of this kind between different studies are at most tentative. The toxicity was significant, but generally manageable. However, with response rates of some 30% and a response duration of around five months there is room for improvement. Combinations of docetaxel and cisplatin with one or more other active agents might further improve the outcome, especially with regard to prolonging response duration, thus helping to palliate patients suffering from the devastating effects of progressive incurable recurrent SCCHN.

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